

Lauren Mai

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Education

Texas A&M University

B.S.E. in Computer Science, Minor in Mathematics; GPA 4.0

College Station, TX

Spring 2025

Relevant Coursework: Programming Languages, Data Structures, Databases, Computer Organization, Design/Analysis of Algorithms, Data Science, Machine Learning, Distributed Systems, Applied Cryptography, Reverse Engineering, Risk Modeling, Parallel Computing

Experience

Bell Flight

Fort Worth, TX

IT Intern – Software Development Engineer

June 2024 – August 2024

- Collaborated with development team on Enterprise Chat Application utilizing *Azure OpenAI API, C#, Blazor Web-assembly, and Cosmos DB* as Fullstack Engineer, supporting 8000+ users.
- Implemented bUnit testing suite for front-end view of Chat Application and addressed backlog of feature development and bug tickets.
- Leveraged Azure DevOps for feature development branches to allow for continuous integration and deployment.

Texas A&M University

College Station, TX

Undergraduate Teaching Assistant – CSCE 120/121

January 2024 – Current

- Conducted 4+ hour weekly office hours every week and assisted 700+ students with technical issues and homework assignments while also offering office hours by appointment.
- Lead lab session for an hour every week, teaching C++ basics and fundamental programming principals and object-oriented concepts to a cohort of 25+ students while also assisting students in utilizing the Linux terminal to compile and run code, promoting their proficiency in using the command line interface.
- Participated in weekly meetings with course staff and sub-teams to discuss course progression and current backlog of assigned tasks, which included grading of student exams, plagiarism detection, and absence documentation.

Texas A&M University

College Station, TX

Undergraduate Research – Hyperspectral-based Feature Extraction Algorithms

August 2023 – May 2024

- Worked in a team to develop semantic segmentation algorithms in *C++ and CUDA* within Docker container Linux environment using multithreading and thread pooling applications for optimization.
- Added Hyperspectral image functionality to pre-existing Visual Odometry project, leveraging OpenCV and feature detection algorithms while improving data processing workflow through use of system calls to automate file parsing.
- Collaborated with team in an agile scrum environment, utilizing Github for version tracking through use of feature development branches, ensuring for successful research presentation at symposium.

Personal Projects

Gong-Cha Web Application ([Website Link](#) and [Demo Link](#))

October – December 2023

- Developed and deployed visually appealing and interactive full-stack application using *Google APIs, OAuth, JavaScript, Express.js, Node.js, and Amazon RDS for PostgreSQL*, allowing managers to perform CRUD operations on the company inventory, cashiers to take and customize customer orders, and customers to dynamically order and customize their own drinks in an online e-commerce format.
- Collaborated within a team environment following agile scrum methodologies using Jira, with weekly standup meetings to ensure communication of project features.
- Utilized cookies to manage user sessions, allowing for prevention of unauthorized access to those with role restricted access based on group identity.

Fault Tolerant, Scalable, and Available Tiny SNS CLI Application

April – May 2024

- Implemented and designed a scalable CLI SNS application through use of gRPC and ProtoBuf protocols in Ubuntu environment through use of VirtualBox, allowing for clients to follow, unfollow, and stream messages in a synchronous manner to followers in real time.
- Leveraged Master-Slave and Coordinator based distributed system to ensure fault-tolerance and high availability of data and ensured liveness of servers through use of keep-alive messages and zNodes for state management and leader election.
- Established dynamic scaling by replication of follower states across multiple servers through use of forwarding state within Master/Slave clusters and cluster follower synchronizer processes, allowing for causal and eventual consistency on user end.

Skills & Interests

Languages: Java, Python, C++, HTML, CSS, LaTeX, JavaScript, Haskell, CUDA, HDL, SQL, C#, Bash, C, x86 ASM, Shell Script, XML

Frameworks/Technologies: NumPy, Matplotlib, Pandas, Artifactory, Google Cloud, Seaborn, Waterfall, Java Swing, TensorFlow, DOM, Visual Studio, Material UI, Jest, Ghidra, xUnit, bUnit, VirtualBox, gRPC, Azure Cosmos Db, Azure DevOps, Blazor, PostgreSQL, Hadoop, Sklearn